SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-015: Student Records Export to CSV

Field	Details
Change Request ID	CR-015
Change Request Title	Student Records Export to CSV
Student Name	AAM. Nusky
Student ID	JF/ICT/24/01
Implementation Status	Implemented; test execution and evidence capture pending
Date	15 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators need to download and use student records outside the Student Management System for administrative, reporting, and record-keeping purposes. Student information was available only inside the system, with no facility to export it into a portable file format. The change adds a secure Export CSV facility so that authorized administrators can download student records as a CSV file that opens in spreadsheet software.

1.2 Existing Behaviour

- The administrator could view student records within the existing student management interface.
- No CSV export facility was available.
- Administrators had to manually view or copy student information to create an external record.
- Student CRUD operations use the existing student identifiers.
- Administrator authentication (JWT) and role authorization already protect student management functionality and must remain unchanged.

1.3 Requested Change

- Provide an Export CSV button for authorized administrators in the student management interface.
- Export Student ID, Registration Number, First Name, Last Name, Email, and Phone Number with CSV column headers.
- Generate a properly formatted CSV file, handle multiple records, and handle an empty student list without error.
- Download the file to the administrator's device using a meaningful filename (students.csv).
- Preserve existing viewing, editing, and deleting functionality and existing authentication and authorization.
- Ensure student records are not exposed to unauthorized users and that exporting does not modify the original data.

2. Implementation

2.1 Implementation Summary

An Export CSV button was added to the administrator student management page (DashboardPage). When clicked, the frontend sends an authenticated GET request to a new protected backend endpoint, GET /api/students/export. The endpoint is mounted behind the existing JWT authentication and administrator role-authorization middleware, so unauthenticated and non-administrator requests are rejected before any student data is read.

The backend retrieves the existing student records using a read-only query, builds the CSV text with a fixed header row and correctly escaped values, and returns it with CSV content-type and attachment headers. The frontend receives the response as a Blob and triggers a browser download named students.csv. If there are no students, a header-only CSV is returned and the interface remains stable. No database structure, authentication rule, or existing CRUD operation was changed.

2.2 Affected Files

File	Change / Purpose
frontend/src/pages/DashboardPage.jsx	Adds the Export CSV button, click handler, loading state, and error message to the student management interface.
frontend/src/services/studentService.js	Adds an exportStudentsCsv request that sends the authenticated API call and receives the file as a Blob.
Dashboard / student management stylesheet	Adds responsive styling for the Export CSV control.
backend/src/routes/studentRoutes.js	Registers GET /api/students/export behind existing authentication and admin authorization middleware, ahead of any /:id route.
backend/src/controllers/studentController.js	Adds the export handler: fetches records, builds the CSV, sets response headers, and returns the file.
backend/src/models/studentModel.js	Adds a read-only query that returns the student fields required for export.

Note: No additional files outside those identified in the change request were required.

2.3 Impact Analysis

Area	Impact	Control / Result
Frontend UI	One new button and handler in the student management page.	Existing table, edit, and delete controls are untouched; the button uses the existing styling approach.
Backend API	One new read-only GET endpoint.	No change to existing endpoints; route placed before parameterized routes to avoid conflicts.
Database	Read-only SELECT of existing student fields.	No schema change, no new table, no data modification.
Authentication	Export reuses the existing JWT verification.	Authentication logic unchanged.
Authorization	Export reuses the existing administrator role check.	Authorization rules unchanged; non-admin users are rejected.
Existing CRUD	Shares the same student model file.	Existing functions retained; only a new function was added.
Performance	Whole student list is read and returned in one response.	Acceptable for the current data size; pagination/streaming is out of scope.

2.4 Technical and Security Changes

CSV Export Implementation

- Endpoint: GET /api/students/export (authenticated, administrator only).
- Columns, in order: Student ID, Registration Number, First Name, Last Name, Email, Phone Number.
- The header row is always written, so the file is valid even when no students exist.
- Each value is converted to text and wrapped in double quotes when it contains a comma, quotation mark, or line break; embedded quotation marks are doubled, following standard CSV rules.
- Null or missing values (for example a student without an email) are written as empty cells.
- Response headers: Content-Type text/csv; charset=utf-8 and Content-Disposition attachment with filename students.csv.
- Frontend: the response is read as a Blob, a temporary object URL is created, a hidden link is clicked to start the download, and the URL is then revoked.

Security Controls

- The route uses the same authentication and administrator authorization middleware as the existing student management routes (SEC-001, SEC-002).
- Unauthenticated requests receive 401 and non-administrator users receive 403; no student data is generated for them (SEC-003).
- The export query selects only the six required fields; password hashes and other account data are never included.
- Spreadsheet formula injection is mitigated: values that begin with =, +, -, or @ are prefixed with an apostrophe so spreadsheet software treats them as text.
- The export is read-only and contains no INSERT, UPDATE, or DELETE statement, so original records cannot be changed (AC-012).
- No authentication or authorization code was modified (SEC-004).

2.5 Implementation Logic

1. The administrator opens the student management page and clicks Export CSV.
2. The frontend disables the button, shows a loading state, and sends GET /api/students/export with the stored JWT.
3. The backend verifies the token and confirms the administrator role; otherwise it returns 401 or 403.
4. The model runs a read-only query for Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.
5. The controller builds the header row and one escaped row for each student; with no students, only the header row is produced.
6. The controller sets the CSV headers and returns the file content.
7. The frontend converts the response to a Blob and downloads it as students.csv.
8. The button is re-enabled and an error message is displayed if the request failed.

2.6 Error and Data Handling

- Empty student list: a header-only CSV is returned and downloaded; no runtime error occurs (FR-008).
- Missing token or invalid token: the request is rejected with an authentication error and the frontend shows a controlled message.
- Non-administrator user: the request is rejected with an authorization error.
- Database or server error: a controlled error response is returned, and the frontend shows an export-failed message without breaking the page.
- Special characters in names or other fields are escaped so that the CSV structure is not corrupted.
- Exported data is read directly from the same source used by the student list, so it reflects the records available in the system (FR-007).

3. Testing and Verification

3.1 Testing Approach

Testing covers the Export CSV interface, the generated file, the API authorization behaviour, empty-data handling, and regression of existing functionality. Frontend behaviour is checked in the browser, the CSV is opened in spreadsheet software, and API authorization is checked with requests made without a token and with a non-administrator token. A before-and-after record count confirms that the export does not modify data. The Actual Result and Status columns are recorded as each test is executed; until then the status remains Pending.

3.2 Test Cases and Results

Test ID	Test Case	Expected Result	Actual Result	Status
TC-001	Log in as administrator and open the student management page	Export CSV button is visible.	To be recorded after execution.	Pending
TC-002	Click Export CSV with student records present	A CSV file is downloaded without error.	To be recorded after execution.	Pending
TC-003	Check the downloaded filename	File is named students.csv.	To be recorded after execution.	Pending
TC-004	Open the CSV and check the header row	Headers: Student ID, Registration Number, First Name, Last Name, Email, Phone Number.	To be recorded after execution.	Pending
TC-005	Compare values for one student with the system record	All six values match the system record.	To be recorded after execution.	Pending
TC-006	Export with multiple students	Every student appears as one row.	To be recorded after execution.	Pending
TC-007	Compare the full CSV with the student list on screen	Exported data matches the displayed records.	To be recorded after execution.	Pending
TC-008	Export with no student records	Header-only CSV is downloaded; no runtime error.	To be recorded after execution.	Pending
TC-009	Export a student whose name or field contains a comma or quotation mark	CSV columns remain correctly aligned in spreadsheet software.	To be recorded after execution.	Pending
TC-010	Call GET /api/students/export without a token	Request is rejected with 401; no data returned.	To be recorded after execution.	Pending
TC-011	Call GET /api/students/export with a student (non-admin) token	Request is rejected with 403; no data returned.	To be recorded after execution.	Pending
TC-012	Compare student record count before and after export	Record count and data are unchanged.	To be recorded after execution.	Pending
TC-013	View and add student records after the change	Existing view and create operations work correctly.	To be recorded after execution.	Pending
TC-014	Edit a student record after the change	Edit works and saves correctly.	To be recorded after execution.	Pending
TC-015	Delete a student record after the change	Delete works correctly.	To be recorded after execution.	Pending
TC-016	Test administrator login and role-protected access	Existing authentication and authorization remain functional.	To be recorded after execution.	Pending

3.3 Test Summary

Test Result	Count / Result
Total defined test cases	16
Passed	0
Pending execution	16
Failed	0
Execution coverage	0 of 16 (0%)
Overall test result	Not yet executed; update this table after testing

4. Acceptance Criteria Verification

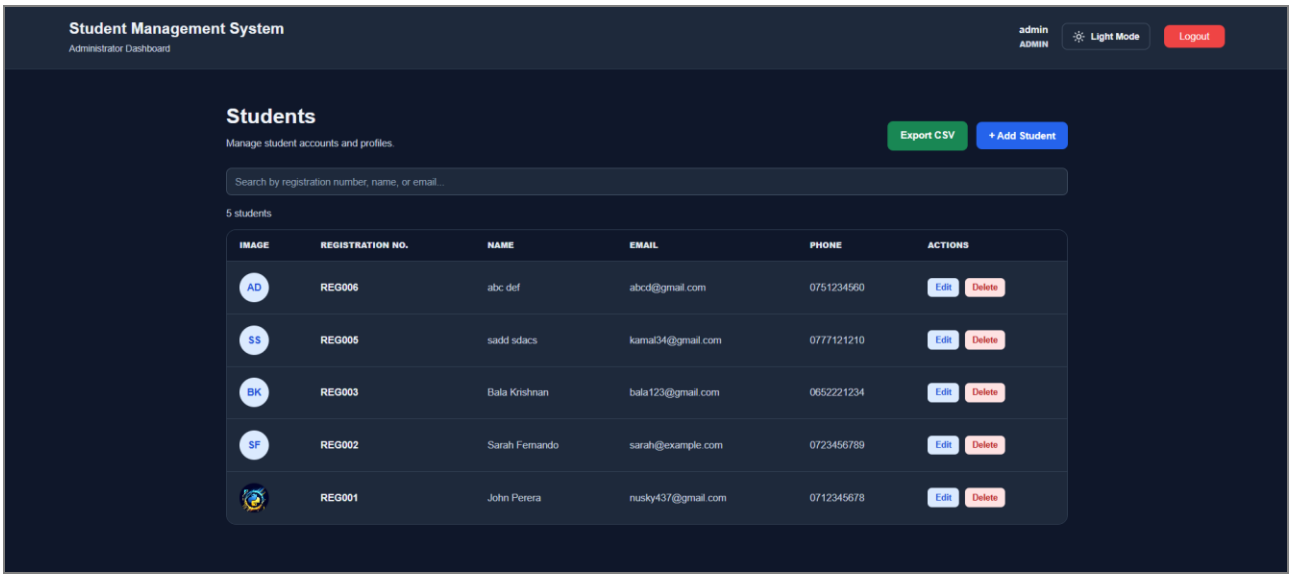
ID	Acceptance Criterion	Implementation / Verification Status
AC-001	Authorized administrator can see the Export CSV option.	Button added to the administrator student management page. Verification: TC-001. Pending.
AC-002	Administrator can download a CSV file with student records.	Export endpoint and Blob download implemented. Verification: TC-002. Pending.
AC-003	CSV contains the required column headers.	Fixed header row implemented. Verification: TC-004. Pending.

AC-004	CSV contains correct Student ID, Registration Number, First Name, Last Name, Email, and Phone Number.	Six fields selected and written in order. Verification: TC-005. Pending.
AC-005	Multiple student records are exported correctly.	One row is written per student. Verification: TC-006. Pending.
AC-006	File has .csv extension and meaningful filename.	Content-Disposition and download name set to students.csv. Verification: TC-003. Pending.
AC-007	Exported data matches the records available in the system.	Data is read from the same student source. Verification: TC-007. Pending.
AC-008	Empty student list is handled without a runtime error.	Header-only CSV returned. Verification: TC-008. Pending.
AC-009	Unauthorized users cannot access the export functionality.	Route protected by authentication and admin authorization middleware. Verification: TC-010, TC-011. Pending.
AC-010	Existing student CRUD functionality continues to work.	CRUD code retained unchanged. Verification: TC-013 to TC-015. Pending.
AC-011	Existing administrator authentication and authorization continue to work.	Authentication and authorization code not modified. Verification: TC-016. Pending.
AC-012	Export does not modify or delete original student records.	Export uses a read-only query only. Verification: TC-012. Pending.

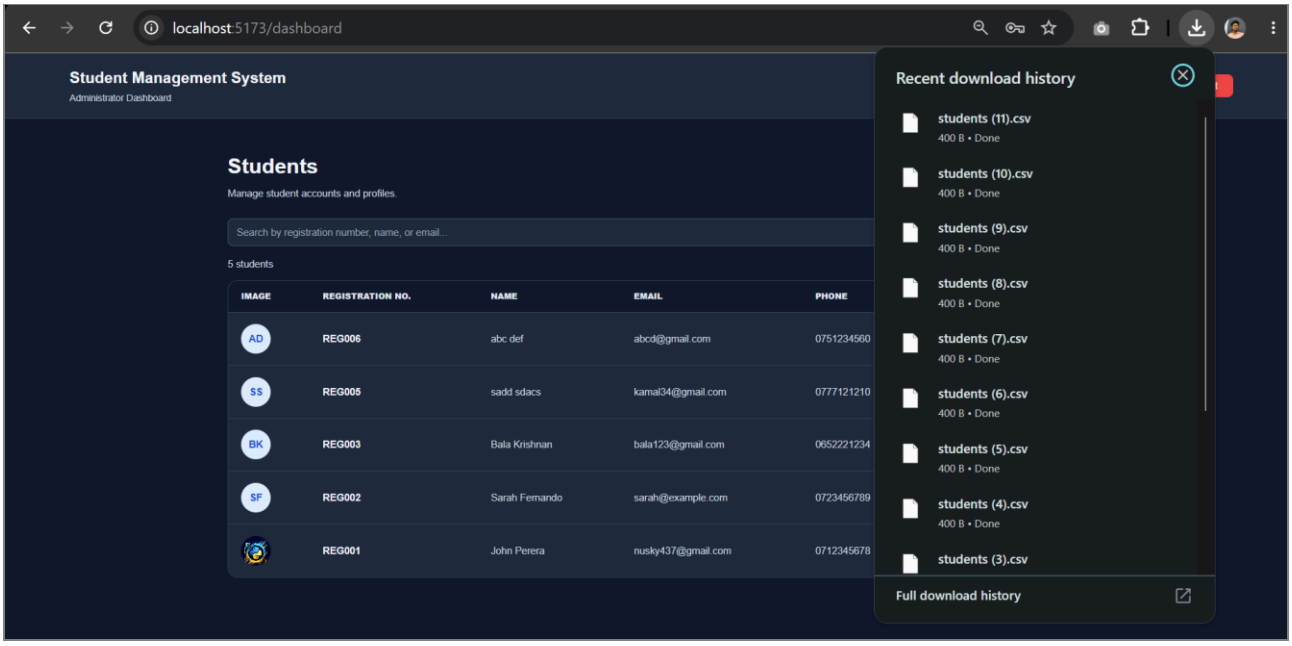
5. Evidence

Screenshots are to be captured during test execution and inserted in the spaces below before formal submission.

Evidence 01 - Student management page showing the Export CSV button.



Evidence 02 - Successful CSV download (browser download notification / downloaded students.csv).



6. Final Result

Item	Result
Requirements	Student records export to CSV has been implemented for authorized administrators.
Implementation	Export CSV button, authenticated export endpoint, read-only data retrieval, CSV generation with escaping, and browser download are present.
Testing	Test cases TC-001 to TC-016 are defined; execution results are pending.
Acceptance criteria	Implementation present for AC-001 to AC-012; verification pending (see section 4).
Final status	Implemented; pending test execution, evidence capture, and formal review.

6.1 Final Summary

CR-015 adds a Student Records Export facility to the administrator student management interface. Authorized administrators can click Export CSV to download students.csv containing Student ID, Registration Number, First Name, Last Name, Email, and Phone Number with column headers. The export is protected by the existing JWT authentication and administrator authorization, uses a read-only query, handles empty lists safely, escapes special characters, and does not change the database, authentication rules, or existing CRUD operations. The implementation should be treated as fully accepted only after the pending test cases and evidence in sections 3 and 5 are completed.

6.2 Issues / Limitations

- Test execution results and screenshots are still to be added before formal submission.
- The export reads all students in one request; for a very large number of records, pagination or streaming would be a future improvement.
- Only CSV format is supported; PDF, XLSX, emailing, cloud storage, and scheduled exports are out of scope for this CR.
- Report and signature are to be completed after the end-to-end tests.

Student Name: AAM. Nusky
Student ID: JF/ICT/24/01

Date: 15 October 2026

Signature: _____

~ END OF REPORT ~